Hassaan Raza

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SUMMARY

Final-year Electrical Engineering student with hands-on experience in industrial automation, embedded systems, and power systems. Experienced in high-voltage grid operations and real-time control architectures. Seeking opportunities in Embedded Systems, Industrial Automation, or AI-driven Engineering roles.

EDUCATION

University of Engineering and Technology, Lahore — Lahore, Pakistan
Bachelor of Electrical Engineering — CGPA: 3.53 — 2022 – 2026

PROFESSIONAL EXPERIENCE

220kV NKLP Grid Station NGCP (Formerly NTDC) — Lahore, Pakistan
Electrical Engineering Intern — Jul 2025 – Sep 2025

National Grid Company Pakistan, dealing with power transmission across Pakistan.

- **Grid Station Operation:** During my internship at NGCP, I was involved in various maintenance activities, including the maintenance of circuit breakers, transformers, and other grid equipment.
- **Power System Protection:** Learned about transmission line, busbar, and transformer protection, along with other protection schemes implemented at the grid station.
- **Fault Troubleshooting:** Worked on troubleshooting a fault in one of the circuit breakers located in the switchyard during my time at the grid station.

ACADEMIC EXPERIENCE & PROJECTS

PLC-Based Automated Bottle Filling, Capping & Vision-Based Defect Detection System
Final Year Project

- **Industrial Automation System:** Leading the design of an industrial automation system integrating PLC (Allen-Bradley MicroLogix 1400) control and computer vision for real-time quality inspection.
- **Bottle Filling & Capping:** Developed an automated bottle filling and capping system with synchronized actuator control using PLC-based logic.
- **Vision-Based Inspection:** Implemented camera-based defect detection to identify errors and product inconsistencies.
- **PLC Communication:** Established Ethernet-based communication between the vision system and PLC for real-time data exchange and decision-making.

Autonomous Collision Detection and Prevention Robot — STM32
Embedded Systems Project

- **Robot Development:** Designed and developed an autonomous robotic vehicle using STM32 microcontroller, ultrasonic sensor, IR sensors, and L298N motor driver.
- **Collision Avoidance:** Implemented real-time obstacle detection and collision avoidance using sensor fusion techniques.

HONORS & AWARDS

- **PEEF Scholarship:** Received full merit-based academic scholarship from PEEF (2017–2019).
- **CARE Foundation Scholarship:** Awarded CARE Foundation scholarship for Intermediate studies (2019–2021).
- **Prime Minister Laptop Award:** Received merit-based Laptop Award under Prime Minister Punjab Scheme 2025.

COURSES

- **Machine Learning Specialization:** DeepLearning.AI (Instructor: Andrew Ng)
- **PLC Fundamental I:** PLC Dojo (Udemy)

TECHNICAL SKILLS

Embedded Systems: STM32, ESP32, CubeIDE, Arduino IDE
Power Systems: Load Flow Analysis, Short Circuit Analysis, Transient Analysis, ETAP, PS World
Automation: RSLogix 500, Process Control, Ladder Logic, CODESYS
Electronics: Proteus, PLECS, KiCad, Power Electronics Design